

Load and Run the Oven Tray SOP (1x Cycle, in situ)

One episode loads one tray, bakes it, and rests it on the trivet at the oven where it lives. The base is **passive**: it has no drive of its own, so it is pushed by hand to the counter in front of the oven and locked there, and nothing is carried away to a table. Everything the episode touches is already there when recording starts: the oven in its cabinet run, the mitt on its hook beside the door, and the trivet, the tray, and the parcel tub on the counter to the right of it.

The episode runs these eight actions in this order and no other: **load the tray, open the door with the mitt, rack the tray, shut the door and set the timer, run the bake, open the door with the mitt and take the tray out, rest the tray on the trivet, shut the door with the mitt.**

The oven is worked **as found**. It is already at temperature and holding when recording starts, the door is shut, the middle rack is empty, the tray is empty on the counter, and the parcel tub is open. The episode ends with the four parcels baked, the tray standing on the trivet, and the door shut.

This is an in-situ task, and four things follow from that. First, there is a **cabinet directly above the oven**, and the cavity has a ceiling of its own, so **every approach is from the front, straight in, level**. No gripper comes down into the cavity from above. Second, the **open door stands out in front of the mouth like a shelf**, and every arm passes over it — no gripper, wrist, or forearm ever rests on it, and nothing is ever set down on it. Third, the **oven is never leaned on and never pushed** — no press on the door, the dial, or the panel is ever hard enough to shift the appliance. Fourth, the **door stands level at its stop on its own**, so neither arm holds it and both arms stay free for the tray.

The oven is hot from the first frame to the last. The cavity, the rack, the door front, the door edge, and the door handle are hot before recording even starts, and the tray metal is hot from the moment it comes out. Only four things are ever touched by a gripper: the **mitt's grip tab**, the **tray's silicone side grips**, the **parcels** while they are still cold on the counter, and the **timer dial**. Nothing else.

The door handle is only ever taken through the mitt. The mitt hangs on its hook beside the door, the **right gripper** takes it by its grip tab for each of the three door moves, and hangs it straight back afterwards. A bare gripper never touches the handle.

Arm roles are fixed and never swap. The **right gripper is the working gripper**: the parcels, the mitt, the door, and the timer dial, and the right side of the tray in both carries. The **left gripper is the holder**: it holds the tray still while the parcels go on, and takes the left side of the tray in both carries.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the counter and stop it **square to the front of the oven**, so the front of the appliance runs straight across the frame and neither end of it sits nearer than the other.
2. Stop it **centered on the door mouth**, so the left side and the right side of the mouth are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach the back of the middle rack straight in through the door mouth without either arm extending, and **far enough** that neither arm, wrist, nor any part of the base touches the oven, the open door, or the cabinet above while both arms work.
4. Check the **height band**: with the base parked, both grippers pass over the open door level, carry the tray onto the middle rack, and the **right gripper** reaches the mitt hook and the timer dial, all without a wrist or forearm fouling the cabinet above.
5. Check the **door swing**: the **right gripper** reaches the door handle with the door shut and can carry it through its whole swing down to its stop without extending, and the open door is clear of both arms' path to the door mouth.
6. Check the **counter**: the **right gripper** reaches into the parcel tub and reaches the tray's right grip, the **left gripper** reaches the tray's left grip, and both grippers reach the trivet, all without extending and without knocking anything.
7. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
8. If any of lines 1 to 6 fails, push the base to a new park by hand and start again at line 1. Do not work an oven the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the cabinet run hard enough to shift it, nothing touches it by hand, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.

2. The environment camera is centered on the door mouth and its frame includes the whole middle rack, the door shut and open, the timer dial, the mitt on its hook, and the trivet, the tray, and the parcel tub on the counter.
3. The camera reads the middle rack from the front, so whether the tray sits square on it and whether it clears the cavity walls are readable.
4. The camera reads the tray from the front, so how many parcels are on it, which quarter each one sits in, and whether any parcel hangs over the tray edge are readable.
5. The camera reads the timer dial, so where the pointer sits is readable.
6. The camera reads the door front once it is shut, so a door standing ajar is readable.
7. Both arms are at home with grippers open.
8. The **right arm** reaches the parcel tub, the tray's right grip, the mitt hook, the door handle with the door shut and with it open, the timer dial, and the back of the middle rack, without extending to a joint limit.
9. The **left arm** reaches the tray's left grip and the back of the middle rack, without extending. It reaches neither the mitt hook, the door handle, nor the timer dial.
10. Both grippers enter and leave the door mouth **from the front and level**, passing over the open door without touching it. Neither comes down into the cavity from above, and neither wrist nor forearm touches the cabinet above on the way in or out.
11. The two arms do not collide while they carry the tray together, and the open door is clear of both arms' path to the door mouth and to the trivet.
12. If a place cannot be reached, re-park the base by the Base positioning steps until lines 8 to 11 hold.

Materials checklist

1. The **oven** stands where it lives, in its cabinet run. It is not moved, not leaned on, and not pushed at any point.
2. There is a **cabinet directly above the oven**, so the only way in is the front-facing door mouth and never the top.
3. The oven is **already at temperature and holding** before recording starts. Bringing it up to temperature is a setup job and is not part of the episode.
4. The **door** is hinged along its bottom edge. It drops forward and down to level and **stands at its stop on its own**, without being held and without drifting.
5. The open door stands out in front of the mouth like a shelf. It is hot, it carries no weight, and nothing is ever set down on it.
6. The **door handle** is a bare metal bar across the top of the door. It is hot, and it is only ever taken through the mitt.
7. The door shuts with a latch that holds it closed. A correctly shut door meets the frame all round, stands flush with the front of the appliance, and does not spring back open.

8. The **timer dial** sits on the panel above the door mouth, clear of the door swing. It is marked in minutes, it turns clockwise, and it rings and returns its pointer to zero when the time runs out.
9. The **middle rack** is in place, sits level, and is empty. The racks above and below it are empty too.
10. The **mitt** hangs on a hook on the cabinet face just to the right of the door, above the counter. It hangs within the **right arm's** reach and is not over the counter or over the trivet.
11. The mitt is a padded pad with a stiff **grip tab** along one edge. The tab holds its shape, so a gripper closed on it holds the mitt flat instead of letting it fold.
12. The mitt is big enough that its padded face covers the whole length of the door handle a gripper takes hold of.
13. The **tray** stands on the counter to the right of the trivet, empty, dry, and flat. It does not rock.
14. The tray is rectangular, with a **flat floor** and a **low raised edge** all round, so where each parcel goes is never a judgment call.
15. The tray has a **wire side handle on each short side**, one on the left and one on the right, each fitted with a **silicone grip**. The grips stay cool enough to be taken by a bare gripper straight out of the oven.
16. The tray is small enough to sit on the middle rack with clear space on both sides, side handles included, so nothing catches a cavity wall going in or coming out.
17. The **trivet** stands on the counter between the oven and the tray. It is heat-proof, it lies flat, it does not slide on the counter, and it is bigger than the tray floor, so the whole tray stands on it.
18. The **parcel tub** stands open on the counter to the right of the tray. It does not slide when a parcel is lifted out of it.
19. The tub holds **four parcels**, all the same size, each wrapped in foil, each with a flat bottom, and each firm enough to keep its shape in a closed gripper.
20. Each parcel is small enough that four of them sit on the tray with clear space between them and none of them over the tray's raised edge.
21. The counter between the trivet, the tray, and the parcel tub is clear, so an arm carrying a parcel or the tray passes over nothing.
22. The counter is deep enough that the tray stands on it clear of the front edge, and the trivet stands clear of the front edge too.
23. Nothing else stands on the counter within either arm's reach.

Workspace layout

Nothing anywhere is marked or taped out. Every place below is judged by eye against the appliance and the cabinet run themselves — the door mouth, the middle rack, and the counter.

- **Oven:** the appliance the base is parked at. It is never moved, never leaned on, and never pushed.
- **Cabinet above:** the cabinet directly over the oven. It is what makes every approach a front approach. Nothing ever touches it.
- **Door mouth:** the open front of the cavity. Both grippers go in and come out through it, level and from the front.
- **Open door:** the dropped door standing level in front of the mouth. Every arm passes over it. Nothing rests on it and nothing is set down on it.
- **Cavity:** the inside of the oven, holding the racks. Nothing goes into it but the tray and the grippers that carry it.
- **Middle rack:** the rack the tray goes on. It is the only rack used.
- **Door handle:** the bar across the top of the door. Taken by the **right gripper** through the mitt, and never bare.
- **Timer dial:** the dial on the panel above the door mouth. **Right gripper only.**
- **Mitt hook:** the hook on the cabinet face just right of the door. The mitt hangs there whenever it is not in use. **Right gripper only.**
- **Trivet:** stands on the counter between the oven and the tray. The tray ends the episode on it.
- **Tray:** stands on the counter to the right of the trivet at the start. Carried by both grippers, one on each silicone side grip.
- **Parcel tub:** the open tub on the counter to the right of the tray. **Right gripper only.**

Arm assignments

- **Right gripper — the working gripper.** Lifts the parcels out of the tub and lays them on the tray, works the mitt, opens and shuts the door through the mitt, turns the timer dial, and takes the tray's right grip in both carries.
- **Left gripper — the holder.** Holds the tray still in Step 1 and takes the tray's left grip in both carries. It never touches the mitt, the door, or the timer dial.
- Nothing is handed between grippers, and only one thing is moved at a time.

Vocabulary

- **Front approach:** the gripper enters and leaves the door mouth level and from the front, passing over the open door without touching it, and never comes down into the cavity from above. Every action at the oven is a front approach.
- **Mitt grip:** the **right gripper** closes on the mitt's stiff grip tab, lays the mitt's padded face down along the top of the door handle, and closes on the handle so the mitt sits between its fingers and the bar. It is the only way the door handle is ever taken hold of.

- **Side-grip hold:** both grippers close on the tray's silicone side grips, the **left gripper** on the left grip and the **right gripper** on the right grip, straight across from each other. It is the only way the tray is ever taken hold of, cold or hot.
- **Quarter:** the tray's flat floor read by eye as four equal squares — back-left, back-right, front-left, and front-right — judged against the tray's own edges. One parcel goes in each.
- **Flat in its quarter:** the parcel sits on its flat bottom in the middle of its quarter, touches no other parcel, and no part of it sits over the tray's raised edge.
- **Square on the rack:** the tray sits flat on the middle rack with the gap between the tray and the cavity wall looking the same on the left as on the right, its back edge clear of the back wall, and its front edge inside the door mouth.
- **At its stop:** the door has dropped all the way forward and stands level on its own, without being held and without drifting, and the whole door mouth is clear.
- **Latched:** the door meets the frame all round, sits flush with the front of the appliance, and does not spring back when the **right gripper** comes off it.
- **Timer set:** the dial pointer sits on the 2 mark and the dial is ticking.
- **Timer ended:** the timer has rung and the pointer sits back on zero.
- **Hot parts:** the cavity, the racks, the door front, the door edge, the door handle, and, from the moment it comes out, the tray's metal. No gripper touches any of them. The mitt's grip tab, the tray's silicone side grips, the parcels on the counter, and the timer dial are the only things a gripper ever touches.
- **Clear of the front:** the arm is drawn back so that no part of it is over the counter, the open door, the panel, or the door mouth.
- **Stands on its own:** after the grippers move clear, the tray stays flat and does not rock, slide, or tip.
- **Fully on the trivet:** the whole tray floor stands over the trivet, with trivet showing on all four sides and no part of the tray down on the bare counter.

Steps

Step 1: Load the tray

Goal: four parcels sit flat, one in each quarter of the tray, and the parcel tub is empty.

1.1 Hold the tray still

- With the **left gripper** closed, set it against the left side of the tray and hold the tray still.
- Hold the tray where it already stands. Do not lift it and do not drag it.
- The **left gripper** stays there until Step 1.4.

Check: the tray stands where it started and does not slide when the **left gripper** settles on it.

1.2 Lay the two back parcels

- With the **right gripper**, close on one parcel in the tub and lift it straight up out of the tub.
- Carry it level to the tray. Do not carry it over the trivet.
- Lower it straight down onto the **back-left quarter** and set it on its flat bottom.
- Open the **right gripper** and draw it straight up off the parcel.
- Do the same with a second parcel onto the **back-right quarter**.
- Load the back quarters first, so no arm ever passes over a parcel already on the tray.

Check: both back parcels are **flat in their quarter**, they do not touch each other, and neither sits over the tray's raised edge.

1.3 Lay the two front parcels

- With the **right gripper**, lift a third parcel out of the tub and set it on the **front-left quarter**, the same way.
- Lift the fourth parcel out of the tub and set it on the **front-right quarter**, the same way.
- Keep every carry clear of the parcels already on the tray.

Check: four parcels are on the tray, one **flat in its quarter** in each of the four quarters, none touching another, none over the raised edge, and the tub is empty. If a parcel is off its quarter or over the edge, take it again with the **right gripper** and set it down flat in its quarter.

1.4 Let go of the tray

- Open the **left gripper** and draw it back off the tray.

Check: the loaded tray stands on its own on the counter and the parcel tub stands where it started.

Expected state: the tray holds four parcels, the tub is empty, the mitt hangs on its hook, and the oven door is still shut.

Step 2: Open the door with the mitt

Goal: the door stands level at its stop, the whole door mouth is clear, and the mitt is back on its hook.

2.1 Take the mitt

- With the **right gripper**, close on the mitt's grip tab and lift the mitt straight off its hook.
- Carry it level to the door. Do not let it hang against the door front on the way.

Check: the **right gripper** holds the mitt by its grip tab and the mitt hangs flat, not folded over on itself.

2.2 Swing the door down to its stop

- With the **right gripper**, take the door handle in a **mitt grip**.

- Draw the handle out toward the front until the latch releases, then lower the door forward until it meets its stop.
- Lower it under control. Do not let it drop, do not jerk it, and do not let it bang open.
- Open the **right gripper** off the handle, keeping hold of the mitt by its grip tab, and draw the arm back clear of the door.

Check: the door stands **at its stop** and stays there on its own, the whole door mouth is clear, and the middle rack is visible front to back. If the door did not run to its stop, take the handle again in a **mitt grip** and lower it the rest of the way.

2.3 Hang the mitt back

- With the **right gripper**, carry the mitt back to its hook and hang it there.
- Open the **right gripper** and draw it clear of the mitt.

Check: the mitt hangs on its hook and hangs flat, and no gripper is touching the oven.

Expected state: the door stands open at its stop, the cavity is empty, the mitt is on its hook, and the loaded tray still stands on the counter.

Step 3: Rack the tray

Goal: the tray sits square on the middle rack and both grippers are back out of the door mouth.

3.1 Take the tray

- Bring both grippers to the tray's silicone side grips: the **left gripper** onto the left grip, the **right gripper** onto the right grip.
- Close both grippers on the grips at the same time, in a **side-grip hold**.
- Do not close on the tray edge, the tray floor, a parcel, or the bare wire of a handle, and do not take the tray with one gripper alone.

Check: both grippers hold the tray by its silicone side grips, straight across from each other, and no parcel has shifted.

3.2 Carry it in and set it on the rack

- Lift the tray straight up off the counter, only high enough to clear it.
- Carry it level to the door mouth, over the open door, and take it straight in from the front. Keep it level the whole way.
- Do not tilt it, do not swing it, and do not let any part of an arm come down on the open door.
- Lower the tray onto the middle rack and set it flat, then push it straight back until its back edge is clear of the back wall.

Check: the tray sits flat on the middle rack, no parcel has slid, and nothing has been knocked.

3.3 Square it and come out

- Open both grippers and, with the **right gripper** closed, nudge the tray until it sits **square on the rack**.
- Nudge it a little at a time. Do not knock it, and do not push it hard enough to move the oven.
- Draw both grippers straight out of the door mouth, level, over the open door.

Check: the gap between the tray and the cavity wall looks the same on the left as on the right, no part of the tray or a side handle touches a wall, and the tray's front edge is inside the door mouth. If the tray sits near a wall, nudge it back with the closed **right gripper**.

Expected state: the loaded tray sits square on the middle rack, the counter holds the trivet and the empty parcel tub, the mitt is on its hook, and the door stands open.

Step 4: Shut the door and set the timer

Goal: the door is latched, the mitt is back on its hook, and the timer is set.

4.1 Shut the door with the mitt

- Check first that both grippers are out of the door mouth and empty.
- With the **right gripper**, close on the mitt's grip tab and lift the mitt off its hook.
- Take the door handle in a **mitt grip** and swing the door up in front of the mouth in one smooth move. Do not swing it fast, and do not let it slam.
- Press the handle straight back with the **mitt grip** until the door is **latched**.
- Press straight back only, and do not press hard enough to move the oven.
- Open the **right gripper** off the handle, keeping hold of the mitt by its grip tab, and draw the arm back off the door.

Check: the door is **latched** — it meets the frame all round, sits flush, and does not spring back when the **right gripper** comes off. If it springs back, press it home once more in a **mitt grip**.

4.2 Hang the mitt back

- With the **right gripper**, carry the mitt back to its hook and hang it there.
- Open the **right gripper** and draw it clear of the mitt.

Check: the mitt hangs on its hook and hangs flat.

4.3 Turn the timer dial

- With the **right gripper** closed, come straight in to the timer dial and close on it.
- Turn the dial **clockwise** until the pointer sits on the **2** mark.
- Turn it in one steady move. Do not turn it back the other way, and do not press it toward the panel.
- Open the **right gripper** and draw it straight back off the dial.

Check: the **timer set** — the pointer sits on the 2 mark and the dial is ticking. If the pointer has gone past 2, leave the dial alone until it ticks down onto the 2 mark, then do not touch it again.

Expected state: the door is latched, the tray is on the middle rack, the timer is set, and the mitt is on its hook.

Step 5: Run the bake

Goal: the bake runs to the end of the timer with both arms clear and still.

5.1 Stand clear while it bakes

- Draw both arms back **clear of the front** and hold them still.
- Touch nothing until the timer has ended: not the door, not the dial, not the mitt, the trivet, the tub, or the counter.
- Wait for the **timer ended** signs: the ring, and the pointer back on zero.

Check: both arms stayed clear and still, and nothing on the counter moved while the bake ran.

Expected state: the timer has ended, the door is still latched, and the loaded tray is hot on the middle rack.

Step 6: Take the tray out

Goal: the hot tray is out of the cavity, held level in a **side-grip hold**, and the door still stands at its stop.

From here to the end of the episode the tray's metal is hot. The grippers touch only the tray's silicone side grips, the mitt's grip tab, and nothing else.

6.1 Open the door with the mitt

- With the **right gripper**, close on the mitt's grip tab and lift the mitt off its hook.
- Take the door handle in a **mitt grip** and lower the door forward to its stop in one smooth move.
- Open the **right gripper** off the handle, keeping hold of the mitt by its grip tab.
- Carry the mitt back to its hook, hang it there, then open the **right gripper** and draw it clear.

Check: the door stands **at its stop** on its own, the tray is still square on the rack, and the mitt hangs on its hook.

6.2 Take the hot tray

- Bring both grippers straight in through the door mouth, level and over the open door, to the tray's silicone side grips.
- Close both grippers on the grips at the same time, in a **side-grip hold**.
- Take hold of the silicone grips only. Do not touch the tray edge, the tray floor, a parcel, the rack, or a cavity wall.

Check: both grippers hold the tray by its silicone side grips, straight across from each other, and nothing else in the cavity has been touched.

6.3 Draw it out

- Lift the tray straight up off the rack, only high enough to clear it.
- Draw it straight out through the door mouth, level and slowly, passing over the open door without touching it.
- Do not tilt it, do not swing it, and do not let it turn between the grippers.

Check: the tray is clear of the door mouth, it is still level, and the four parcels are still one to a quarter.

Expected state: the hot tray is out of the cavity and held level, the cavity is empty, and the door stands open at its stop.

Step 7: Rest the tray on the trivet

Goal: the tray stands flat and fully on the trivet, and both grippers are off it.

7.1 Set the tray down

- Carry the tray level to the trivet. Do not carry it over the parcel tub.
- Hold it level over the trivet, then lower it straight down until it sits flat on the trivet.
- Lower it straight down only. Do not drag it across the trivet, and do not set it down on a corner.
- Open both grippers and draw them back off the silicone side grips.

Check: the tray **stands on its own** and is **fully on the trivet**, with trivet showing on all four sides, and no parcel has slid off its quarter. If any part of the tray is down on the bare counter, take the silicone side grips again in a **side-grip hold**, lift the tray straight up, and set it down again on the trivet.

Expected state: the hot tray stands on the trivet, the cavity is empty, and the door still stands open at its stop.

Step 8: Shut the door with the mitt

Goal: the door is latched and the mitt is back on its hook.

8.1 Swing the door up and latch it

- Check first that both grippers are empty and clear of the tray.
- With the **right gripper**, close on the mitt's grip tab and lift the mitt off its hook.
- Take the door handle in a **mitt grip** and swing the door up in front of the mouth in one smooth move. Do not swing it fast, and do not let it slam.

- Press the handle straight back with the **mitt grip** until the door is **latched**, pressing only as hard as the latch needs.
- Open the **right gripper** off the handle, keeping hold of the mitt by its grip tab, and draw the arm back off the door.

Check: the door is **latched** — it meets the frame all round, sits flush, and does not spring back when the **right gripper** comes off. If it springs back, press it home once more in a **mitt grip**.

8.2 Hang the mitt back

- With the **right gripper**, carry the mitt back to its hook and hang it there.
- Open the **right gripper** and draw it clear of the mitt.

Check: the mitt hangs on its hook, hangs flat, and no part of it lies on the counter or against the oven.

Expected state: the door is shut and latched, the tray stands on the trivet, and the mitt is on its hook.

Step 9: End the episode

- Confirm the tray: it stands flat and fully on the trivet, with four parcels, one in each quarter.
- Confirm the oven: the door is shut and latched and the cavity is empty.
- Confirm the counter: the mitt hangs on its hook, the parcel tub stands empty where it started, and the tray's old spot is clear.
- Confirm nothing has been dropped on the counter, on the open door, or on the floor.
- Confirm the base has not moved and the oven has not shifted.
- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Let the tray cool on the trivet by hand until it can be handled.
2. Take the four parcels off the tray by hand and put them back in the parcel tub, or replace them with four fresh parcels.
3. Wash and dry the tray, then stand it back on the counter to the right of the trivet, empty and flat.
4. Wipe the trivet clean and dry and stand it back on the counter between the oven and the tray.
5. Wipe up anything spilled on the counter, the open door, the rack, or the cavity floor, and dry it.

6. Check the four parcels are still the same size, still firm, and still wrapped, and replace any that have split, gone soft, or lost their flat bottom.
7. Check the mitt: it hangs flat on its hook, its grip tab still holds its shape, and its padded face is not scorched, worn thin, or damp. Replace it if it is.
8. Check the tray's silicone side grips are still fitted, not loose, and not melted, and replace the tray if they are.
9. Replace a tray that has warped, that rocks on the counter, or that no longer sits flat on the middle rack.
10. Check the door still latches and does not spring back when it is pushed shut by hand, and that it still stands level at its stop on its own.
11. Check the middle rack is seated and level, and that the racks above and below it are empty.
12. Check the timer dial turns freely, rings, and returns its pointer to zero.
13. Check the oven is back at temperature and holding before the next episode.
14. Check the base is still locked and parked square, then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the oven shifts in frame, the door mouth changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 9, the base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Approached from above or fouled the cabinet above

- **Visible cue:** a gripper comes down into the cavity from above instead of entering level from the front, or a wrist or forearm knocks, scrapes, or rests on the cabinet above or the top of the door mouth on the way in or out.
- **SOP rule broken:** Steps 3 and 6, every action at the oven is a front approach, level, in and out through the door mouth.
- **Coaching note:** straight in from the front, straight out the same way. There is no way in from the top.

Violation: Rested on or loaded the open door

- **Visible cue:** a gripper, wrist, or forearm comes down on the open door, or the tray, a parcel, or the mitt is set down or rested on it at any point.
- **SOP rule broken:** Steps 3, 6, and 7, every arm passes over the open door without touching it, and nothing is ever set down on it.
- **Coaching note:** the door is not a shelf. Pass over it, never onto it.

Violation: Leaned on or pushed the oven

- **Visible cue:** a gripper, wrist, or forearm rests on the door front, the door mouth, or the panel; the oven rocks or shifts; or a press on the handle or the timer dial moves the appliance.
- **SOP rule broken:** Steps 2 to 8, the oven carries no weight and is never pushed out of place.
- **Coaching note:** the arm holds itself up. Press only as hard as the latch or the dial needs.

Violation: Tray loaded wrong

- **Visible cue:** the **right gripper** puts fewer or more than four parcels on the tray, puts two in one quarter, leaves a quarter empty, leaves parcels touching each other, or leaves a parcel over the tray's raised edge.
- **SOP rule broken:** Steps 1.2 and 1.3, lay one parcel flat in each of the four quarters, none touching another and none over the raised edge.
- **Coaching note:** four parcels, four quarters, one each, all clear of the edge.

Violation: Parcels loaded in the wrong order

- **Visible cue:** the **right gripper** loads a front quarter before both back quarters are filled, or carries a parcel over a parcel already on the tray.
- **SOP rule broken:** Steps 1.2 and 1.3, load the two back quarters first, then the two front quarters, so no arm passes over a parcel already on the tray.
- **Coaching note:** back row first. Nothing ever travels over something already placed.

Violation: Parcel handled wrong

- **Visible cue:** the **right gripper** drags a parcel across the counter or the tray, drops it onto the tray instead of setting it down, squashes it, tips it off its flat bottom, or the **left gripper** lifts a parcel out of the tub.
- **SOP rule broken:** Steps 1.2 and 1.3, the **right gripper** lifts each parcel straight up out of the tub, carries it level, and lowers it straight down onto its flat bottom.
- **Coaching note:** straight up, level across, straight down. Right gripper only.

Violation: Tray not held still while it is loaded

- **Visible cue:** the **left gripper** is not on the tray while parcels are laid on it, comes off it early, or the tray slides, turns, or rocks on the counter while it is loaded.
- **SOP rule broken:** Steps 1.1 and 1.4, the **left gripper** holds the tray still from before the first parcel until after the fourth is placed.
- **Coaching note:** hold it first, then load it. A tray that slides mid-load puts parcels in the wrong quarters.

Violation: Door handle taken without the mitt

- **Visible cue:** a bare gripper closes on the door handle, or a gripper takes the handle while the mitt hangs from the same gripper without its padded face lying between the fingers and the bar.
- **SOP rule broken:** Steps 2.2, 4.1, 6.1, and 8.1, the door handle is only ever taken in a **mitt grip**, with the mitt between the fingers and the bar.
- **Coaching note:** mitt on the bar first, then close. The handle is as hot as the oven.

Violation: Mitt handled or parked wrong

- **Visible cue:** the **right gripper** closes on the mitt's padded face instead of its grip tab; the mitt folds over on itself, is dragged, or is dropped; the **left gripper** touches the mitt; or the mitt is left on the counter, on the oven, or on the tray instead of back on its hook after a door move.
- **SOP rule broken:** Steps 2.1, 2.3, 4.1, 4.2, 6.1, 8.1, and 8.2, the **right gripper** takes the mitt by its grip tab and hangs it straight back on its hook after each door move.
- **Coaching note:** tab to take it, hook to leave it, every single time.

Violation: Door opened wrong or not to its stop

- **Visible cue:** the **left gripper** opens the door; the **right gripper** takes the door somewhere other than its handle; the door is jerked or let fall so it bangs open; the door is held by an arm instead of standing at its stop; or it is left part way down so the mouth is not clear.
- **SOP rule broken:** Steps 2.2 and 6.1, the **right gripper** takes the handle in a **mitt grip** and lowers the door to its stop under control, and the door stands there on its own.
- **Coaching note:** right gripper, handle, controlled drop, all the way to the stop, then let go.

Violation: Tray taken with the wrong grip

- **Visible cue:** one gripper alone lifts the tray, a gripper closes on the tray edge, the tray floor, a parcel, or the bare wire of a handle instead of the silicone side grip, or the two grippers close at different times.
- **SOP rule broken:** Steps 3.1, 6.2, and 7.1, both grippers close on the tray's silicone side grips at the same time, one on each side, straight across from each other.
- **Coaching note:** both hands, both silicone grips, closing together. A one-gripper tray tips.

Violation: Tray tilted, swung, or moved fast

- **Visible cue:** the tray tips out of level during a carry, swings between the grippers, turns between them, is snatched off the rack, or a parcel slides while it is carried, going in or coming out.
- **SOP rule broken:** Steps 3.2, 6.3, and 7.1, carry the tray level and straight, lifting only high enough to clear the surface and drawing it straight out of the door mouth slowly.
- **Coaching note:** level, low, and slow, both ways. A hot tray that tips cannot be caught.

Violation: Tray racked wrong

- **Visible cue:** the tray goes on a rack other than the middle one, sits nearer one cavity wall than the other, touches a wall with its edge or a side handle, sits crooked or not flat on the rack, or sticks out past the door mouth when the grippers come off.
- **SOP rule broken:** Steps 3.2 and 3.3, set the tray flat on the middle rack, push it back clear of the back wall, and nudge it until it is **square on the rack**.
- **Coaching note:** middle rack, flat, pushed back, even gaps. Square it before you come out.

Violation: Door not shut and latched

- **Visible cue:** the door springs back after the **right gripper** comes off, stands ajar or proud of the front, or is swung shut fast enough to slam; the timer dial is turned while the door still stands open or unlatched; or the episode ends with the door at its stop, ajar, or off its latch.
- **SOP rule broken:** Steps 4.1, 4.3, and 8.1, swing the door up and press it home in a **mitt grip** until it is latched, before the dial is turned in Step 4 and before the episode ends in Step 8.
- **Coaching note:** door first, dial second, and door shut before you finish. Watch it settle before you move away.

Violation: Timer set wrong

- **Visible cue:** the **right gripper** leaves the pointer on a mark other than 2, turns the dial counter-clockwise, turns it back after overshooting, presses it toward the panel, or the **left gripper** touches the dial.

- **SOP rule broken:** Step 4.3, the **right gripper** turns the dial clockwise in one steady move until the pointer sits on the 2 mark.
- **Coaching note:** clockwise, one move, stop on 2. If you overshoot, let it tick down and leave it alone.

Violation: Arms not clear or something touched while the bake ran

- **Visible cue:** an arm stays over the counter, the open door, the panel, or the door mouth while the bake runs, an arm moves during the bake, or a gripper touches the door, the dial, the mitt, the trivet, or the tub before the timer has ended.
- **SOP rule broken:** Step 5.1, draw both arms back clear of the front and hold them still, touching nothing until the timer has ended.
- **Coaching note:** hands off while it bakes. The bake is part of the episode, not a break in it.

Violation: Door opened before the timer ended

- **Visible cue:** the **right gripper** takes the mitt or the door handle while the dial is still ticking and its pointer is still off zero, or before the timer has rung.
- **SOP rule broken:** Step 6.1, wait for the ring and the pointer back on zero before the door is opened.
- **Coaching note:** wait for the ring. Opening early ends the bake short and throws heat at the arms.

Violation: Hot part touched after the bake

- **Visible cue:** after the bake, a gripper touches the tray edge, the tray floor, a parcel, the rack, a cavity wall, the door front, or the door edge, or an arm reaches into the cavity for anything other than the tray's silicone side grips.
- **SOP rule broken:** Steps 6.1 to 8.2, from the end of the bake the grippers touch only the tray's silicone side grips and the mitt's grip tab.
- **Coaching note:** two things only. If it was in the oven, it is hot.

Violation: Tray set down off the trivet or set down wrong

- **Visible cue:** the tray is set down with part of it on the bare counter, set down on a corner, dragged across the trivet, dropped the last part of the way, let go before it sits flat, or it rocks or slides after the grippers come off.
- **SOP rule broken:** Step 7.1, lower the tray straight down until it stands flat and **fully on the trivet**, and only then open both grippers.
- **Coaching note:** all the way down, flat, trivet showing all round, then let go.

Violation: Wrong order of work

- **Visible cue:** the door is opened before the tray is loaded, the tray goes in before the door is open, the timer is set before the door is latched, the tray comes out before the timer has

ended, or the door is shut before the tray is on the trivet.

- **SOP rule broken:** Steps 1 to 8, load the tray, open the door, rack the tray, shut the door and set the timer, run the bake, take the tray out, rest it on the trivet, shut the door.
- **Coaching note:** the order is the task. Every step leaves the station ready for the next one.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped, or a check is made and the fault it finds is left uncorrected — a parcel off its quarter or over the edge, a tray not square on the rack, a door that springs back, a pointer off the 2 mark, a tray part off the trivet, or a rocking tray.
- **SOP rule broken:** Steps 1.1 to 8.2, run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, spilled, or knocked over

- **Visible cue:** the tray, a parcel, or the mitt is dropped on the counter, on the open door, in the cavity, or on the floor; the parcel tub or the trivet is knocked out of place or tipped; or the tray is knocked off the rack.
- **SOP rule broken:** Steps 1 to 8, nothing is dropped, spilled, or knocked out of its place, and every gripper leaves the door mouth straight out to the front.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** takes a parcel, the mitt, the door handle, or the timer dial; the **right gripper** holds the tray still while the other works; the tray is taken by one gripper alone; or anything is handed from one gripper to the other.
- **SOP rule broken:** Steps 1 to 8, the **right gripper** works the parcels, the mitt, the door, and the dial, the **left gripper** holds the tray, and both take the silicone side grips together for the two carries.
- **Coaching note:** right arm works, left arm holds, both arms carry the tray, and nothing changes hands.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the tray still in the cavity or still on the counter, a parcel off the tray, the door open or unlatched, the mitt off its hook, something left on the open door, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 9, confirm the tray, the oven, the counter, the drops, and the base, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read the middle rack, the tray, or the timer dial**, so whether the tray was square, which quarter each parcel went in, or where the pointer sits cannot be judged.
- **Hardware fault on an arm**: gripper failure, drift, controller caused collision, or motor error.
- **Base fault**: a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Oven off temperature**: an oven that has not held its temperature, or that loses it during the bake.
- **Faulty door**: a door whose latch will not hold, that will not stand level at its stop, that drifts, or that falls open instead of lowering under control.
- **Faulty timer**: a dial that will not turn, that does not tick down, that does not ring, or that does not return its pointer to zero.
- **Faulty rack**: a middle rack that is not seated, that is not level, or that tips a correctly placed tray.
- **Door mouth too tight**: a mouth so low or so narrow that both grippers cannot enter level from the front holding the tray, or cannot reach the back of the middle rack.
- **Faulty mitt**: a mitt whose grip tab loses its shape or comes off in a correct grip, whose padding lets heat through, or that will not hang on its hook.
- **Defective tray**: one that warps in the bake, that rocks on the counter or the trivet, whose silicone side grips come loose, melt, or get too hot to take bare, or that slips out of a correct side-grip hold.
- **Trivet will not stay put**: a trivet that slides on the counter when a tray is set down on it, or that is too small for the tray to stand fully on.
- **Parcels will not behave**: parcels that split, sag, or lose their flat bottom in a correct grip, or that will not stay put in their quarter during a correct carry.
- **Parcel tub will not stay put**: a tub that slides or tips on the counter when a parcel is lifted out of it.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the parcel tub, the tray's grips, the mitt hook, the door handle, the timer dial, the trivet, or the back of the middle rack cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Hold the tray still on the counter

2. Lift one parcel out of the tub
3. Set one parcel down in its quarter of the tray
4. Lift the mitt off its hook
5. Take the door handle in a mitt grip
6. Lower the oven door to its stop
7. Hang the mitt back on its hook
8. Take the tray by both silicone side grips
9. Carry the tray into the cavity and set it on the middle rack
10. Nudge the tray square on the rack
11. Swing the oven door up and press it home
12. Turn the timer dial to the mark
13. Stand clear while the bake runs
14. Draw the tray out of the cavity
15. Set the tray down on the trivet
16. Return both arms home and end the episode